

Ecological Data Visualization as a Means to Narrative Building

Kiara C Cushway, U.S. Army Engineer Research and Development Center, Vicksburg, MS, United States and UIC Government Services, Bowhead Family of Companies, Springfield, VA, United States

Jasmine Rodriguez, Department of Biology, Texas State University, San Marcos, TX, United States

Iris R Foxfoot, U.S. Army Engineer Research and Development Center, Vicksburg, MS, United States; UIC Government Services, Bowhead Family of Companies, Springfield, VA, United States; and Sierra Nevada Conservancy, Auburn, CA, United States

S Kyle McKay, U.S. Army Engineer Research and Development Center, Vicksburg, MS, United States and Woolpert, New York, NY, United States

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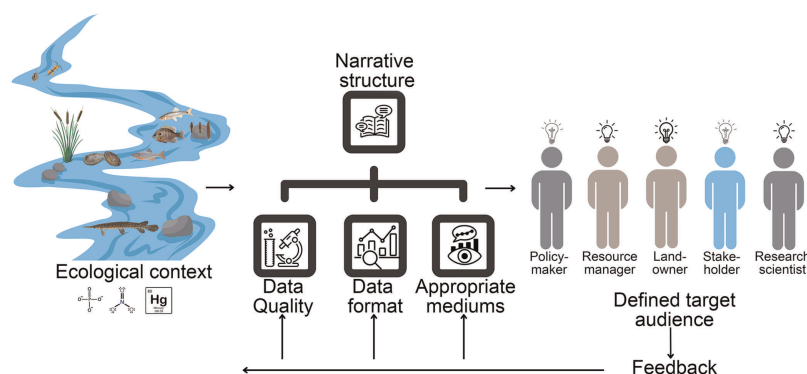
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Abstract

Data visualization is a key component of scientific exploration, explanation, and communication. Here, we present an overview of data visualization as a means of narrative building to communicate ecological research to various technical and non-technical audiences. We explore important considerations for visualizing ecological data, including identifying a target audience, deciding on a visualization method, and ensuring accessibility. We demonstrate these concepts with examples from the Texas Research Institute for Aquatic and Groundwater Ecology (TRIAGE).

Graphical Abstract



Key Points

- Data visualization is a key skill for ecologists looking to explore, explain, or communicate data.
- Narrative building can help ecologists communicate complex data and ideas to varied audiences.
- Effective communication and visualization are centered around the target audience.
- Quality data visualization requires consideration of medium, visualization method, and accessibility.

Introduction

Ecologists increasingly encounter large datasets associated with ever-maturing data collection platforms (e.g., remote sensing, sensor modernization), novel simulation methods (e.g., forecasting and modeling tools), and growing communities of engaged stakeholders (e.g., citizen scientists). This era of “big data” (Hampton *et al.*, 2013) introduces unique opportunities for insight as well as challenges for analysis and interpretation. Information visualization provides a useful means to explore, synthesize, and communicate about ecological processes and outcomes (McInerney *et al.*, 2014). In particular, ecological data visualization draws from the fields of visual analytics, information visualization, computer graphics, and scientific communication (McCormick *et al.*, 1987; Keim *et al.*, 2008; Tufte, 2001; Aigner *et al.*, 2023) to inform ecological analysis, inquiry, and decision-making (McKay, 2019; Hadjimichael *et al.*, 2024).

Data visualization methods can be brought to bear on a variety of issues facing ecologists, ranging from examining findings of basic research to communicating those findings to non-scientific audiences. However, selection of the “right” visualization can be challenging for multiple reasons. First, the purpose and goals of visualization are context-dependent, and visualization can be used to inform analysis and insight (i.e., exploratory purposes) as well as communicate outcomes or inform decisions (i.e., explanatory purposes) (Hadjimichael *et al.*, 2024). Second, visualization methods must be tailored to a target audience’s level of knowledge, background, information needs, and data literacy (Kress *et al.*, 2021). Third, the multitude of data visualization and plotting methods provide an overwhelming set of options to consider. Fourth, seemingly small design choices about visualizations can both tap into cognitive skills (Healey and Enns, 2012) and potentially introduce cognitive biases (Ellis, 2018).

In this chapter, we provide a qualitative guide to key components of data visualization in ecology (Fig. 1). Rather than undertake a foolhardy review of the multitude of visualization methods, we emphasize the importance of data visualization as one component of narrative building. We begin by discussing the role of narratives in basic and applied ecology. Then, we emphasize the importance of audience-centric development of visualizations. A set of considerations are subsequently presented to guide ecologists through selection and use of visualization. We conclude with a set of freshwater ecology case studies demonstrating these concepts.

Narrative Building in Ecology

The complexity of linking theoretical concepts, empirical data, and management implications is a fundamental challenge in ecology (Grainger *et al.*, 2016). While communication challenges in this field are well-recognized (Perra and Brinkman, 2021; Heidenreich *et al.*, 2025), exponential growth of ecological data (Farley *et al.*, 2018) adds further complexity to these communication barriers. Traditionally, scientific communication has focused on audiences already familiar with ecological concepts (e.g., sharing findings in scientific journals or at conferences). However, effective environmental communication is also critical with non-research audiences ranging from policymakers to stakeholders, which requires adapting current communication strategies to various backgrounds and ways of knowing. Perceived values from different communication styles, conflicting narratives, and personal orientations to nature further complicate science communication (Comer *et al.*, 2014; Dahlstrom, 2014; Vigliano Relva and Jung, 2021). Effective ecological communication benefits from intersections with other domains of scientific

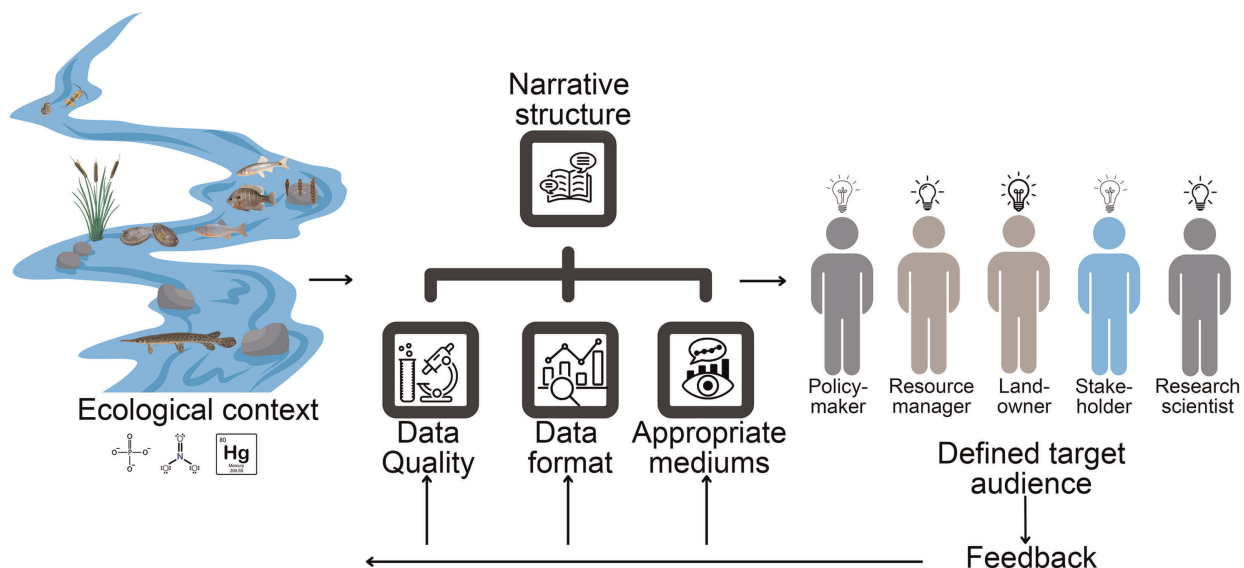


Fig. 1 Key components of effective data visualization.

discourse (Enquist *et al.*, 2017), including the psychology of human reasoning, the philosophy of human-environment relationships, and the sociology of cultural and political influences on human perspectives (Pidgeon and Fischhoff, 2011; Hall *et al.*, 2014; Morris *et al.*, 2019). Increasingly, narrative building has been recognized as a method for integrating these domains and bridging the gap between ecological knowledge and effective communication (Figueiras, 2014).

Storytelling is an evolutionary and critical means of educating, entertaining, and maintaining social relationships (Smith *et al.*, 2017). Incorporating narrative structure and providing the ‘who, what, when, where, how, and why’ establishes connections between events or concepts, provides context that aids in interpretation, and increases knowledge retention and audience curiosity about ecological research (Segel and Heer, 2010). While narrative building as a means of storytelling has been recognized and structurally examined for data visualization in the media (*e.g.*, martini-glass structure, interactive slideshow, and drill down story narrative visualization formats) (Segel and Heer, 2010; Figueiras, 2014), comparable narrative structures and visualization remain under-utilized in ecology. However, the potential of narrative building is increasingly recognized as a means of communicating scientific concepts when addressing topics such as global change, conservation, and ecological theory (Moezzi *et al.*, 2017; Otto and Rosales, 2020). Following a narrative structure has been shown to not only increase engagement among target audiences (*e.g.*, environmental managers, policymakers) but also encourage audience engagement that may otherwise be hesitantly received through traditional means of science communication (Dahlstrom, 2014; Toomey, 2023).

Narratives tend to follow a structured approach when effectively transforming data into a story. Jones and Crow (2017) used an empirical approach called the narrative policy framework when measuring efficacy of narratives in policymaking. The setting, characters, plot, and moral of a narrative, as well as their interplay, can influence the audience’s reaction towards specific concepts, which can improve the memorability of the story (Jones and Crow, 2017). Although emotional connection in scientific communication can seem paradoxical, understanding and consideration of identity-defining characteristics (*e.g.*, cultural, religious, political beliefs) can change the outcome of how individuals and societies process narratives and associated visualizations. Moreover, intended outcomes from narrative building need to be considered when the focus is to persuade or mobilize audiences. Persuasive narratives intend for audiences to change their mind or perception of a topic, while mobilization aims to draw from existing aligned beliefs and their prioritization (Jones and Crow, 2017).

Narrative structures are widely employed across academic disciplines, but their application within western research methods and the ecological literature remains comparatively limited. In contrast, use of narrative communication and visualization of ecology has been practiced for generations by Indigenous peoples, serving as a crucial component in their cultural practices and connection with the land (Cajete, 1997; Datta, 2018). Traditional ecological knowledge is an interconnected part of Indigenous culture and lifeway that strengthens spiritual connection with the land, in which expert knowledge of local ecosystems is conveyed via “storytelling involving vocal and body expression, facial animation, context, plot, character development, intonation, and use of voice” (Datta, 2018, pg. 3). Notably, traditional knowledge sources and communication methods vary across communities and should be recognized for their uniqueness (Datta, 2018; Robinson *et al.* 2021).

Ultimately, storytelling and narrative building has contributed to science communication and ecological knowledge from ancestral, Indigenous, and scientific methods. Visualizations can provide a key means to forming narratives (Tufte, 2001). Maps, diagrams, and figures do more than present information; they transform raw data into interpretable images that tell a story, allowing complex ecological topics to become meaningful, comprehensive, and accessible. Visual structuring mechanisms can guide an audience through an image’s components using techniques such as visual highlighting (*e.g.*, color, size, orientation, motion, *etc.*), annotation (*e.g.*, numbered steps), color-coding, or specific structuring techniques (*e.g.*, left-to-right progression) (Fig. 2) (Segel and Heer, 2010; Healey and Enns, 2012). As a result, visualizations can be understood as narrative structures, influencing how ecological knowledge is developed, interpreted, communicated, and shared with different audiences (Sarkar *et al.*, 2024).

Developing an effective narrative involves balancing the goals of the communicator and the audience’s capacity for engagement, absorption, and retention (Dahlstrom, 2014; Sundin *et al.*, 2018). Narrative building is an iterative process that begins with identifying the message and target audience, establishing a connection between the narrator and those receiving the information (Downs, 2014; Perra and Brinkman, 2021). Establishing a common understanding between the narrator and audience can address initial engagement challenges by ensuring the narrative is conveyed in an appropriate and relatable voice (Norris *et al.*, 2005; Downs, 2014). Additionally, establishing this common ground can help build trust between the narrator and the audience. To sustain that trust, narratives should consider potential sources of bias and be grounded in peer-reviewed research (Downs, 2014). Once a connection with the audience is established, constructing ecological narratives along with corresponding “user-centered” visualizations can further allow audiences to make meaningful connections and increase retention of information (Perra and Brinkman, 2021).

Fig. 2 illustrates an example of an ecological narrative depicting the lifecycle of freshwater mussels. First, ecological data visualization needs to have clearly articulated purpose in terms of why communication is needed. In this case, highlighting this ecological relationship can engage the audience in understanding the role of fish populations in mussel conservation (Modesto *et al.*, 2018). Second, the narrative needs to include what data, information, findings, or knowledge are being conveyed (*i.e.*, lifecycle of mussels). Third, all stories require a setting (*e.g.*, a riverbed, as in Fig. 2), and where a visualization takes place gives the ecological context of a given study, management action, or geography. A clear description of each of these topics provides a foundation for narrative building that continues with identifying an audience and appropriate visualization methods.

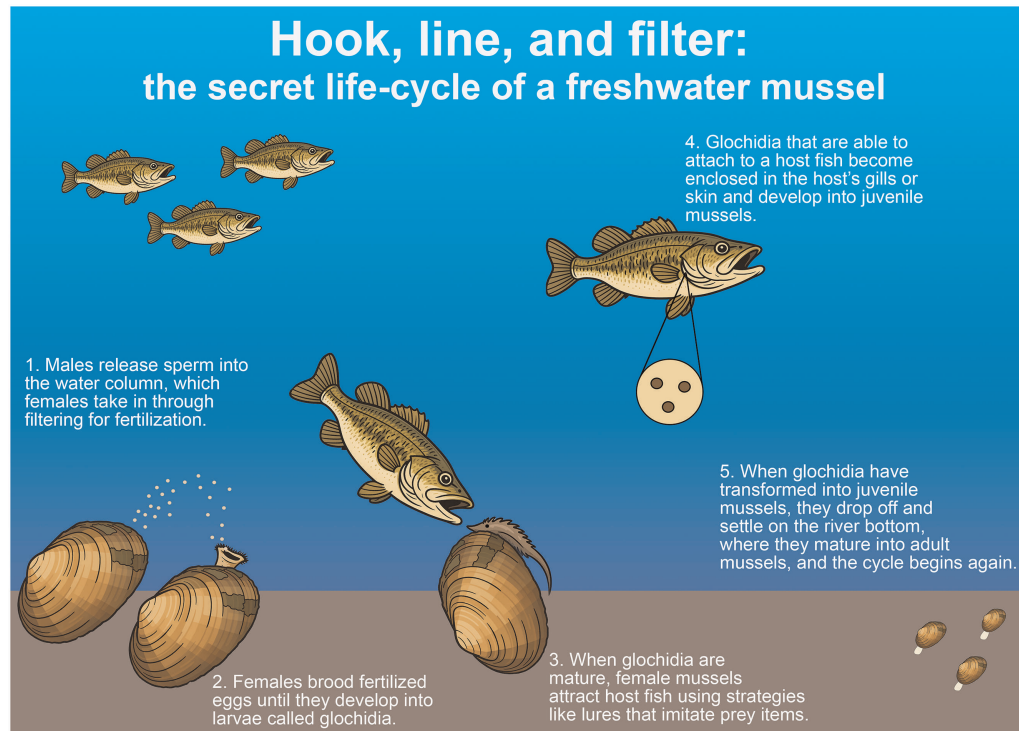


Fig. 2 Example of narrative visualization demonstrating the lifecycle of freshwater mussels. Numbered text and a left-to-right structure guide the viewer through the 'story' of how mussels reproduce with the help of host fish.

Effective Communication Focuses on the Audience (not the Storyteller)

Identification of a target audience is a key aspect of all forms of science communication and is particularly so for designing effective visualizations. The role of stakeholders in ecology and natural resource management is increasingly recognized as a key component of successful policymaking and effective research (Reed *et al.*, 2009). However, communicating ecological ideas to audiences or stakeholders with varied backgrounds can be challenging. Different audiences interact with and understand ecological systems in different ways that are shaped by historical, linguistic, and cultural context (Coscieme *et al.*, 2020). Data visualization as a language for communicating ecological concepts must therefore account for the individual, social, and institutional background of intended audiences.

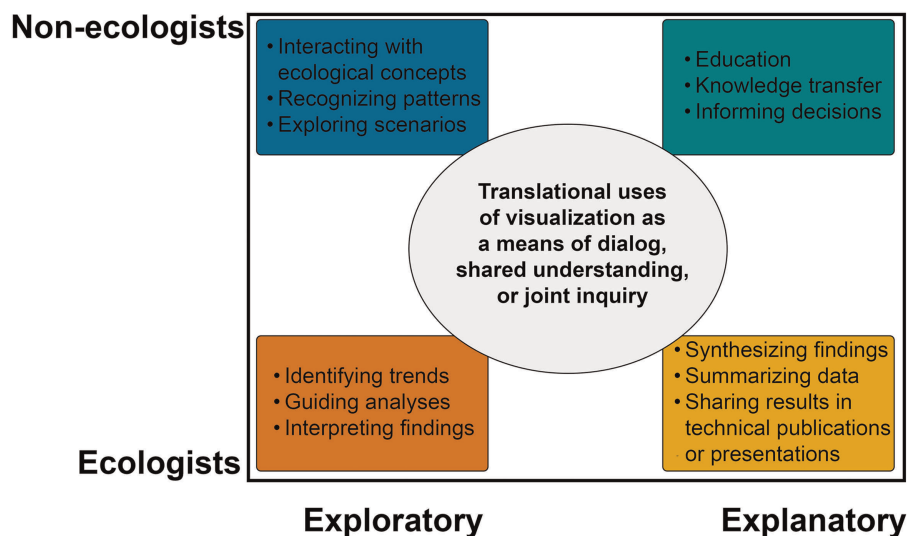
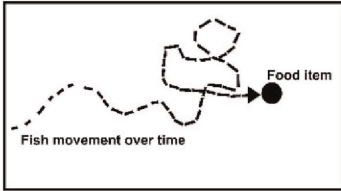
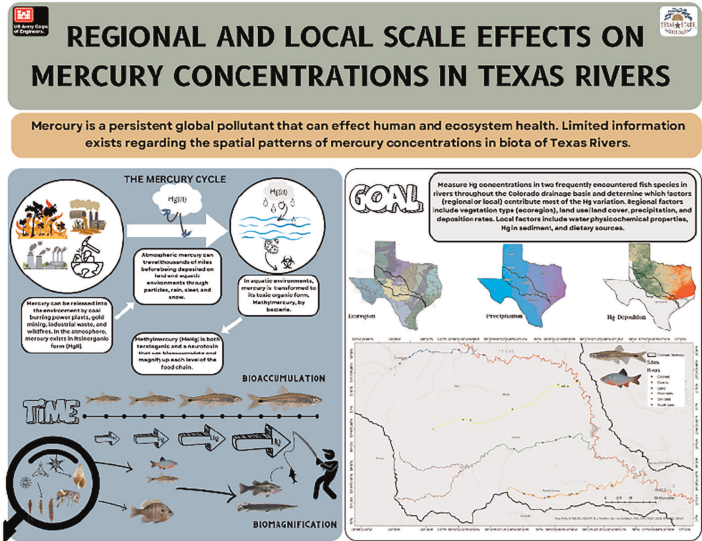


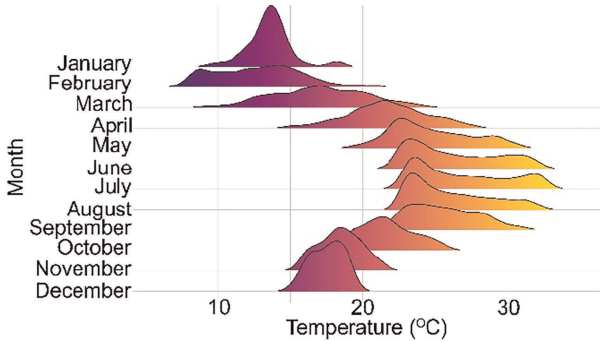
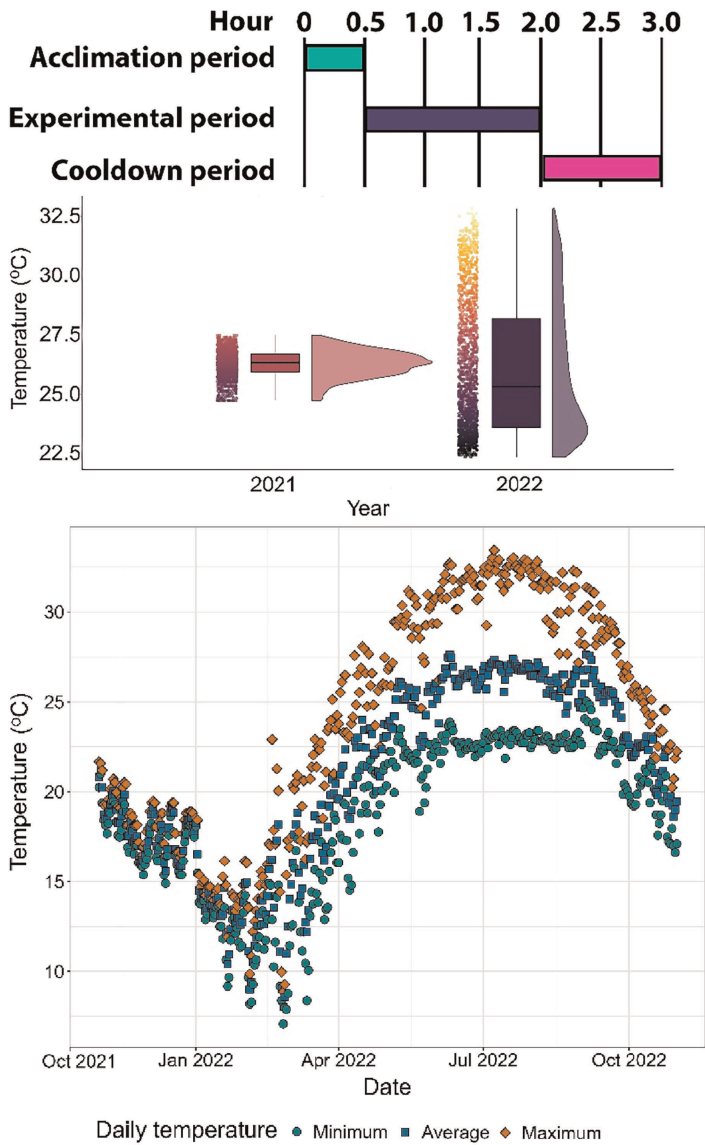
Fig. 3 Translational uses of visualization for exploring and explaining data to diverse audiences.

Table 1 Examples of potential visualization methods often applied to different information types. Note that this is not a complete list of visualizations, and other resources such as those included in the relevant websites list located at the end of this chapter may provide useful starting points for options.

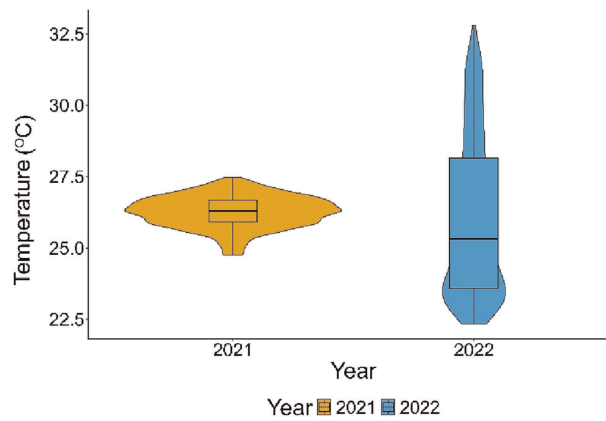
Information type	Possible visualization methods	Examples
Singular events	• Drawings • Illustrations • Photos • Videos	 
Systems, models, or methods	• Conceptual models • Schematics • Infographics	 REGIONAL AND LOCAL SCALE EFFECTS ON MERCURY CONCENTRATIONS IN TEXAS RIVERS Mercury is a persistent global pollutant that can effect human and ecosystem health. Limited information exists regarding the spatial patterns of mercury concentrations in biota of Texas Rivers. THE MERCURY CYCLE Mercury can be released into the environment by coal-burning power plants, gold mining, industrial waste, and wildfires. In the atmosphere, mercury exists in its gaseous form (Hg⁰). Atmospheric mercury can be transformed into reactive forms (methylmercury, Hg²⁺) by natural processes, including soil, land and aquatic environments through particles, rain, sleet, and snow. In aquatic environments, mercury is transformed to its highly toxic form, methylmercury, by bacteria. Methylmercury (MeHg) is both neurotoxic and a carcinogen. It can bioaccumulate and magnify up each level of the food chain. BIOACCUMULATION BIOAMPLIFICATION Measure Hg concentrations in two frequently encountered fish species in rivers throughout the Colorado drainage basin and determine which factors (regional or local) contribute most of the Hg variation. Regional factors include vegetation type (ecoregion), land use/land cover, precipitation, and deposition rates. Local factors include water physicochemical properties, Hg in sediment, and dietary sources. Edward's Plateau ecoregion Average precipitation (mm) 1981-2010 Maximum temperature (°C) 30-year normal 0 250 500 Kilometers Source: U.S. Environmental Protection Agency, 2013 Source: U.S. Department of Agriculture, National Conservation Resource Service, 2014 Copyright ©2025, PRISM Group, Oregon State University, https://prism.oregonstate.edu/Map created 8/12/2025.
Spatial data	• Maps • Choropleths • Raster images • Paths or trajectories	
Time series	• Line plots • Timelines • Ridgeline plots • Gantt charts • Heat maps • Spiral diagrams	

(Continued)

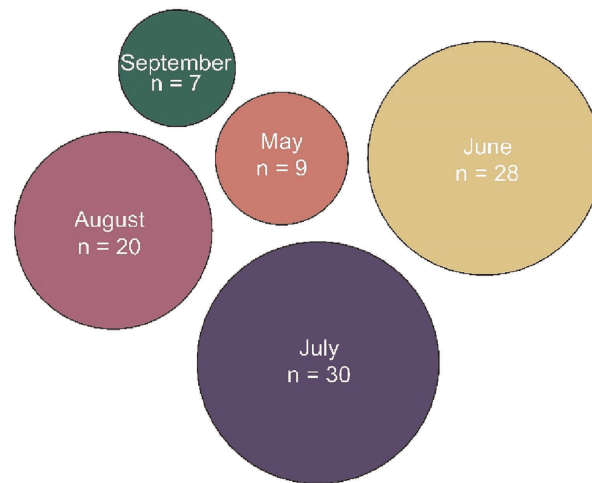
Table 1 (Continued)

Information type	Possible visualization methods	Examples
		
Numerical data	• Scatterplots • Line plots • Raincloud plots • Histograms • Violin plots	
Categorical data	• Bar graphs • Pie charts • Boxplots • Heat maps	

- Waffle charts
- Circle packing charts
- Raincloud plots
- Violin plots



Days per month over 30 °C, summer 2022
Site 106, San Saba River, TX, USA



A target audience can be identified using stakeholder analysis, which helps define who influences or is influenced by a particular visualization (Reed *et al.*, 2009). Individuals or groups that have a stake (*i.e.*, interest and/or influence) in ecological information may have varying levels of understanding, interest, and power to investigate or apply their knowledge (Reed *et al.*, 2009). Tailoring visualizations to specific stakeholders or audiences requires understanding the constraints on their ability to interpret and retain knowledge through different visualization methods. Furthermore, the purpose of a visualization often differs based on the target audience, so identifying an audience can help articulate the underlying goals of visualization (Fig. 3).

When attempting to identify a target audience, ecologists should consider factors like the scope of their research (*i.e.*, who may be affected by or interested in the information), audience demographics (*e.g.*, age, educational background), and end-user goals (*i.e.*, is the visualization purely educational, or will it inform policy or management?) (Böttinger *et al.*, 2020). This information can be used to influence design choices like the level of detail in a visualization, the visualization medium, or the level of interactivity (Böttinger *et al.*, 2020).

Deciding on a Visualization Method

Once a narrative has been crafted and an intended audience identified, decisions must be made about how best to convey the ideas visually. This requires careful consideration of the medium of the visualization (*e.g.*, print, digital, animated), the type of visual method, the interpretability of the message, and accessibility of the figure.

The publication medium is a key aspect of reaching and engaging a target audience. For example, for scientists and policymakers, digital publications are often readily accessible, whereas social media or other non-traditional methods of science communication may be more accessible to members of the public (Bowman and Bozich, 2015). This decision is important early

in the visualization process because the publication medium will dictate how the visualization should be rendered. For example, visualizations for social media should be viewable on a mobile phone, and print visualizations need to be understandable as static images. There is no one-size-fits-all distribution mechanism, and the publication outlet should be carefully considered for a given target audience. For instance, while digital or social media-based outreach may effectively reach younger audiences, it may not be accessible to others. For example, [Baker et al. \(2024\)](#) found that elderly or non-English speaking members of the fire-prone community of Piru, CA, preferred information about wildfire risk via physical signs and flyers in public places.

The next consideration is how to design a visualization that will be easily interpreted by the intended audience. The type of information being conveyed is a key determinant of the appropriate visualization method. Although the types of data visualizations are too numerous to list comprehensively, [Table 1](#) summarizes some overarching visualization methods for conveying particular information types. It is often constructive to mock-up, prototype, or draw potential visualizations with hypothetical data before committing to a particular type of visualization ([Midway, 2020](#)). A given data set may also require multiple visualizations and dialog with collaborators or the proposed audience to home in on the most effective method.

In addition to the form of visualization, factors like color, size, depth, luminance, *etc.* can emphasize key data points or trends and facilitate associations (*e.g.*, darker red colors indicating greater drought severity; [Fig. 4](#)) ([Healey and Enns, 2012](#); [Midway, 2020](#)). Legends, captions, axis labels, and data sources are all critical components of data visualization that provide context and guides for the audience ([Midway, 2020](#)). Text may be used minimally but strategically to reinforce specific parts of a visualization, for example by labeling a peak flow in a hydrograph with a named storm or highlighting a well-known drought period ([Fig. 4](#)).

The accessibility of a visualization should then be considered relative to the audience's data literacy, familiarity with the subject matter, and visual processing capabilities. For example, colorblindness is an often-cited impediment to visual processing that can hinder an individual's ability to interpret visualizations that use certain combinations of colors ([Fig. 5](#)). Colorblind friendly color palettes or integration of patterns or symbols can enhance interpretability in the absence of color cues ([Osiobe et al., 2024](#)). Several common visualization platforms¹ now offer proofing tools that test the interpretability of images relative to different forms of colorblindness. Accessibility may also be improved by including larger, easily readable fonts and using alternative image captioning ([Osiobe et al., 2024](#)). Accessibility issues can also arise in situations involving audiences with lower literacy rates or non-native speakers of the dominant language (see [Baker et al., 2024](#)). Use of jargon or scientific concepts can also reduce understandability of an image without appropriate context or background. In general, the Web Content Accessibility Guidelines (see relevant websites list located at the end of the chapter) are a useful mechanism for proofing visualizations and ensuring they reach the target audience.

After a figure has been conceptualized, it must be rendered. Tools for rendering visualizations can range from pencil and paper to coding software, depending on the requirements of the visualization, time and resources, and abilities of the author. Common rendering tools in ecology include Adobe Illustrator, Photoshop, and Microsoft PowerPoint for conceptual information, Excel, MATLAB, R, or Python for quantitative data, and Geographic Information Systems like ArcGIS and QGIS for spatial data. Increasingly, generative AI may assist with conceptualizing figures, selecting color palettes, and aiding with code-based visualization, although user oversight is needed to ensure accuracy.

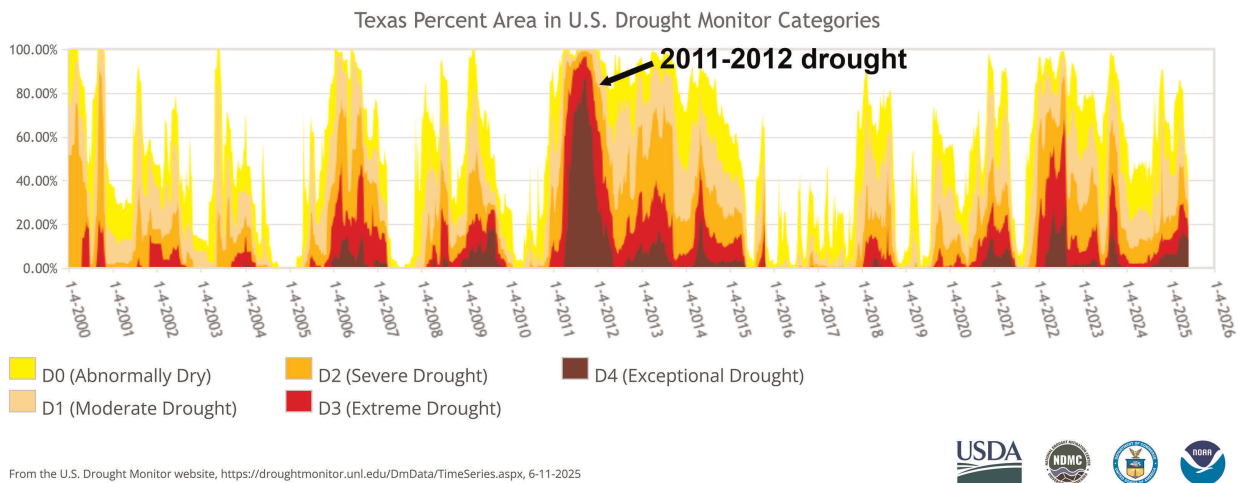


Fig. 4 Example of strategic use of color and text to emphasize trends in drought severity in Texas, USA from 2000 to 2025. The graph displays the percent of Texas' total area falling in particular drought categories over time, with darker shades indicating more severe drought conditions. The 2011–2012 drought, which is considered one of the most severe droughts in Texas history ([Nielsen-Gammon, 2012](#)), is called out using text to draw the viewer's attention. The U.S. Drought Monitor is jointly produced by the National Drought Mitigation Center (NDMC) at the University of Nebraska-Lincoln, the United States Department of Agriculture, and the National Oceanic and Atmospheric Administration. Figure courtesy of NDMC.

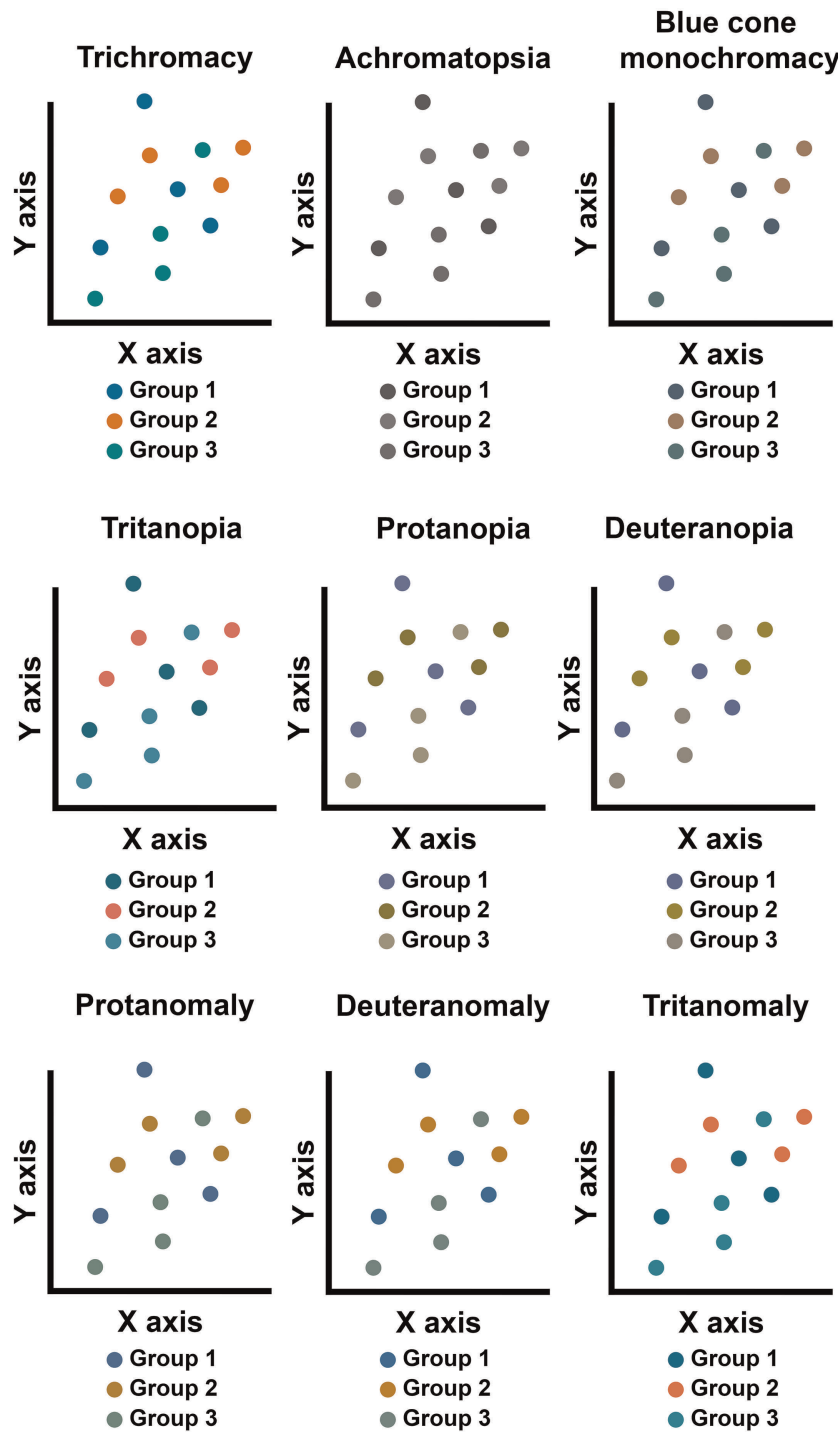


Fig. 5 Examples of how individuals with different types of color vision deficiency perceive colors in a scatterplot. Individuals with trichromacy see the broadest spectrum of colors. Achromatopsia represents total colorblindness, while blue cone monochromacy occurs when red and green cones do not function properly. Tritanopia represents a form of blue-yellow colorblindness, while protanopia and deuteranopia represent forms of red-green colorblindness, and protanomaly and deuteranomaly represent reduced sensitivity to red or green light, respectively. Tritanomaly represents reduced sensitivity to blue light.

Once rendered, feedback from unbiased sources can significantly improve visualizations. Collaborators or colleagues can provide helpful tips when the target audience is within the author's discipline, but it is often equally useful to have someone less

¹ 'Color Proofing' tool in the Adobe Suite, the 'Color Vision Simulator' in ArcGIS Pro, the 'colorblindr' package in R

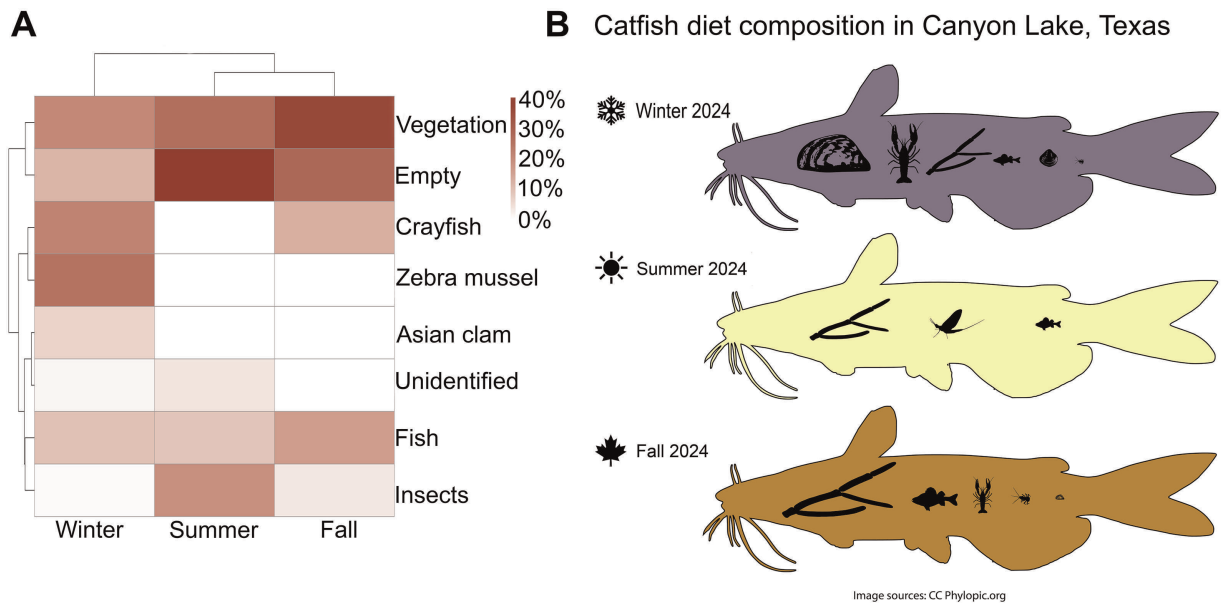


Fig. 6 Methods of visualizing differences in seasonal gut contents of catfish (family Ictaluridae) in Canyon Lake, Texas across the winter, summer, and fall seasons of 2024 using two approaches: A) a heat map of results for a technical audience and B) a graphic depicting scaled images of food resources for a more general audience. Figure adapted from original work by Sarah Stannard (Stannard, 2025).

familiar with a subject critique the interpretability of the visualization (Grainger *et al.*, 2016). Ideally, the author can get feedback from a member of the target audience. The author should be open to criticism and expect to make several iterations of the visualization based on feedback.

Tailoring Visualizations to Heterogeneous Audiences: A Case Study of the Texas Research Institute for Aquatic and Groundwater Ecology (TRIAGE)

When viewed through a narrative and audience-centric lens, data visualization requires a great degree of customization from application to application. As a result, case studies in visualization choices provide a useful tool for describing how visualizations are developed. The Texas Research Institute for Aquatic and Groundwater Ecology (TRIAGE) is an interdisciplinary collaboration between academic institutions and the U.S. Army Corps of Engineers including dozens of research teams. TRIAGE conducts research in freshwater ecosystems that supports coupling of field data with cutting-edge numerical modeling to inform ecosystem management. Since the TRIAGE team includes biologists, ecologists, hydrologists, hydrogeologists, data scientists, and engineers, effective data visualization is an important tool for communicating not only across these varied backgrounds but also with communities where the research is conducted.

Communicating With Technical Audiences: Data Visualization for Scientific Outlets

Clear and effective data visualization is essential for communicating research findings across the various TRIAGE teams, and two of the most common forms of communication with technical audiences are scientific publications and presentations. Written and oral outlets differ in many ways, affecting data visualization choices such as technical depth of content (Midway, 2020), constraints on static vs. dynamic visuals (Hegarty, 2004; Bucher and Niemann, 2012), the amount of time an audience is spending with the content, and flexibility for creative expression, to name a few. Scientific publications and presentations commonly rely on data visualizations that emphasize statistical outcomes, such as heatmaps. However, visuals that synthesize and communicate overarching patterns may be more effective for audiences with differing levels of expertise. Examples of each are outlined below.

In 2024, TRIAGE researchers conducted a study in collaboration with the Texas Parks and Wildlife Department on the prevalence of Zebra Mussels (*Dreissena polymorpha*) in the gut contents of catfish (family Ictaluridae) in Canyon Lake, Texas (Stannard, 2025). The study found that catfish diets varied seasonally, with more than a quarter of their diet composed of Zebra Mussels in the winter, whereas consumption of algae and vegetation was more prevalent in summer and fall. For scientific publication, a heat map was used to visualize seasonal trends in diet switching (Fig. 6A). This figure accurately conveys the results to the technically narrow audience of a discipline-specific journal alongside the statistical similarity of the alternative food sources. However, an alternative visualization (Fig. 6B) was developed for an oral presentation to the interdisciplinary TRIAGE

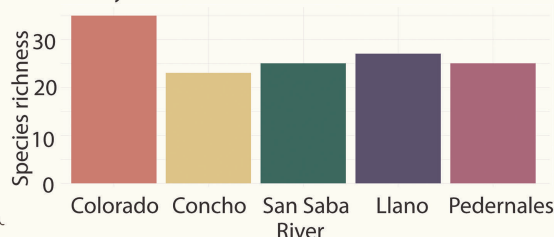
ECOLOGY ALONG THE COLORADO RIVER

In 2021, our team at the Texas Research Institute for Aquatic and Groundwater Ecology (TRIAGE) collected biological data at 100 sites throughout the upper Colorado River and four of its tributaries. Our goal was to understand how the many organisms that call the river their home are distributed and to begin long-term monitoring to monitor future changes.



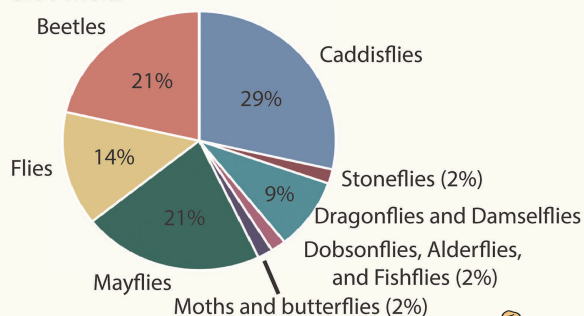
Fishes

We documented 42 different fish species across the 5 rivers. The Red Shiner had the highest average density across all rivers.



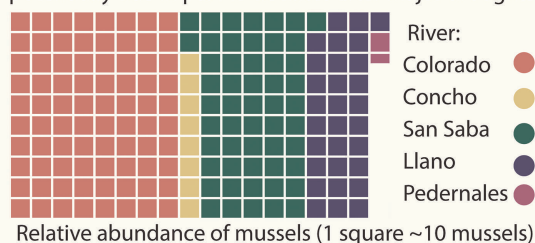
Aquatic insects

There are many types of insects that live in the water for all or part of their lives. We collected at least 33 different families belonging to 8 distinct orders across the 5 rivers.



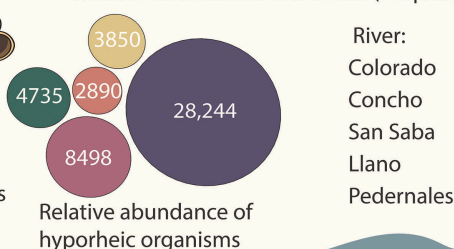
Mussels

We documented 1725 mussels of 15 species across the 5 rivers, including 3 federally endangered species! Mussels were absent at many sites, which was potentially due in part to declines after major droughts.



Hyporheic zone

The hyporheic zone is the area where the surface water and the groundwater meet, and is home to many important river processes and organisms. We collected data in the hyporheic zone at a subset of 40 of the 100 sites, with more than 48,000 specimens and 28 orders of hyporheic organisms collected.



Learn more about our efforts here: <https://www.triagetexas.org/>

Fig. 7 Example of an infographic appropriate for sharing ecological data with the public.

team. **Fig. 6B** scales each of the alternative food sources to reflect their relative proportion of the diet by season. While statistical outcomes are excluded, the audience is given the overarching finding of the work, which can be visually processed quickly in the span of a presentation and provides a more general, non-technical view of the results. In both cases, the visuals were true to the overarching study findings (i.e., the narrative was consistent) but tailored to the constraints of the medium and audience.

Reaching Out: Data Visualization for Public Consumption

Visualizations intended for scientists may differ considerably from visualizations that target the public or policymakers. Use of overly complex techniques or discipline-specific jargon can limit the interpretability of visualizations for non-scientific audiences. Hence, tailoring visualizations to a target audience is an important aspect of creating effective outreach materials. Below, we outline an example using field data collected in Texas.

In 2021, TRIAGE researchers conducted extensive field surveys in the upper Colorado River basin in central Texas (for more information, see [Perez Rocha et al., 2023](#)). Surveys were carried out across 100 sites in five rivers along several gradients of climate and ecology. Researchers collected data on fish, mussel, algae, and macroinvertebrate communities at each site along with water quality and hydrological characteristics. These data are of interest to assorted audiences ranging from researchers to water managers to private landowners. The TRIAGE team developed general outreach materials for these audiences with the goals of spreading situational awareness about the unique data set, sharing high-level findings about local ecosystems, and thanking the stakeholders providing river access for data collection. These stakeholders vary in their degrees of ecological and local knowledge as well as their potential uses of the data. An infographic format (**Fig. 7**) was used to succinctly visualize some of the data, educate audiences about local ecological resources, and point to more detailed information on the TRIAGE homepage. Within the overarching infographic, multiple visualizations were tailored to the particular structures of the individual datasets, but united through consistent color-palettes and levels of detail. The infographic provided a mechanism for creating situational awareness across a wide swath of interested parties and summarizing outcomes from a complex, multi-interdisciplinary research endeavor.

Conclusions

Data visualization is an increasingly important tool in the field of ecology due to larger than ever quantities of data, which must be explored and converted into interpretable narratives. Additionally, it is becoming more apparent that ecology must be accessible in order to meaningfully inform the public dialog around management of the natural world, necessitating visualization strategies that are able to communicate messages across various audiences with differing technical backgrounds. To achieve these objectives, ecologists must thoughtfully consider what narrative they are attempting to convey via data visualization and who the desired audience is. Rather than a comprehensive review of the types of data visualization, this chapter has provided a guide for navigating some of the decisions that ecologists must consider when crafting a successful data visualization.

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Relevant Websites

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Periodic table of visualization methods.

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Information is beautiful.

<https://browser.timeviz.net/>
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<https://r-graph-gallery.com/>
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TRIAGE.